



基于领域自适应的 工业过程故障诊断方法研究

面向 TEP 数据集的无监督领域自适应故障诊断

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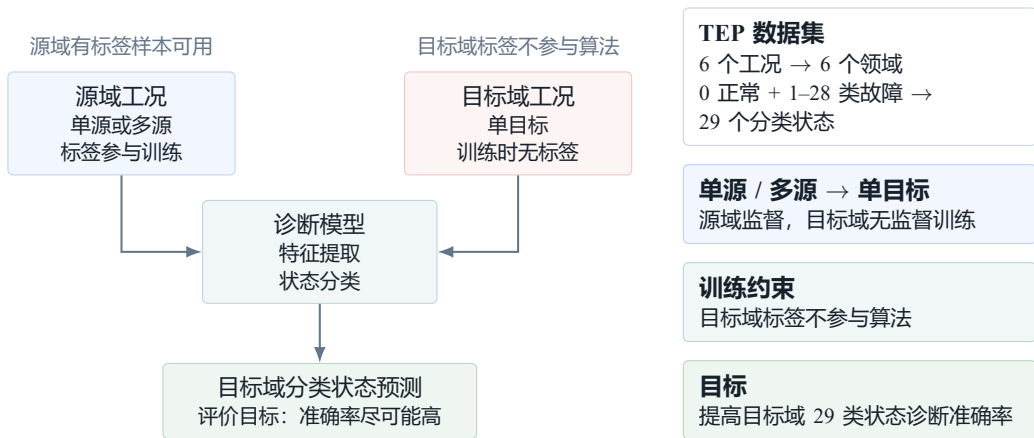
自动化学院

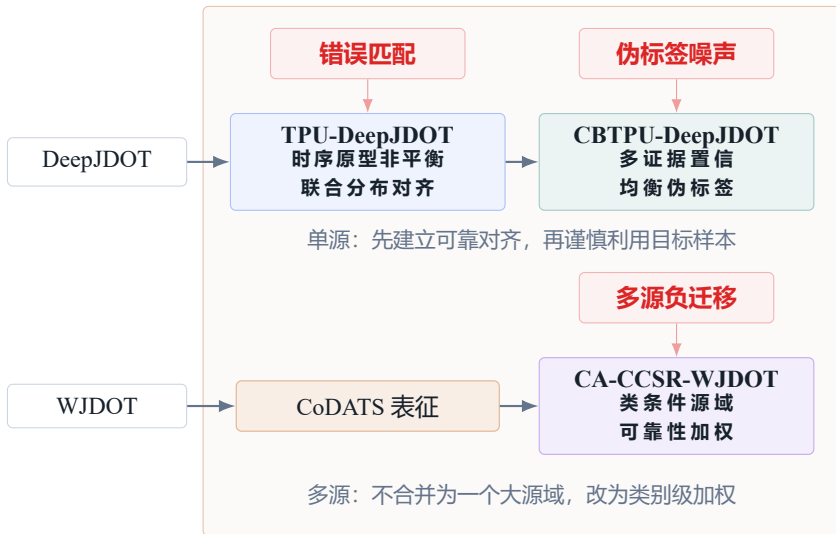
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研究问题界定：TE 过程无监督领域自适应故障诊断

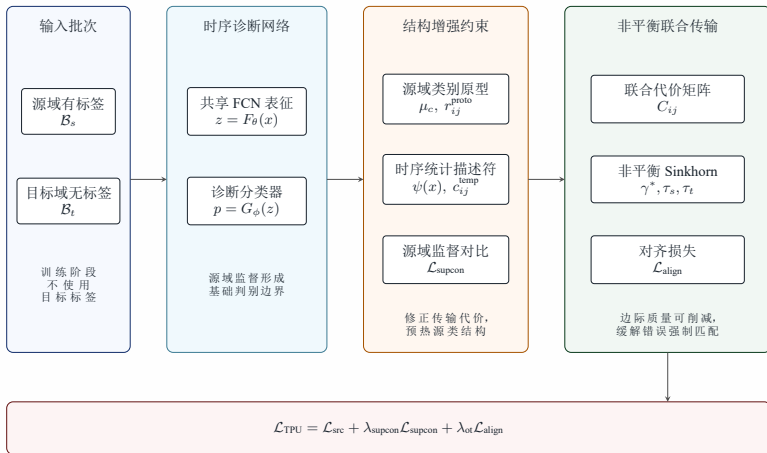
P40

聚焦 TE 过程及其数据集，将领域自适应故障诊断定义为无监督跨工况自适应问题。





TPU-DeepJDOT 时序原型非平衡联合分布对齐机制



联合代价

$$C_{ij} = \lambda_f c_{ij}^f + \lambda_y c_{ij}^y + \lambda_p(\nu) r_{ij}^{\text{proto}} + \lambda_{\text{temp}}(\nu) c_{ij}^{\text{temp}} \quad (3-13)$$

原型惩罚

$$r_{ij}^{\text{proto}} = \text{clip} \left(\frac{d_{y_i^s j}^{\text{proto}} - \bar{d}_{-y_i^s, j}^{\text{proto}}}{\sigma_j^{\text{proto}} + \epsilon}, r_{\min}, r_{\max} \right) \quad (3-7)$$

时序代价

$$c_{ij}^{\text{temp}} = \frac{1}{d_\psi} \|\psi(x_i^s) - \psi(x_j^t)\|_2^2 \quad (3-11)$$

总目标

$$\mathcal{L}_{\text{TPU}} = \mathcal{L}_{\text{src}} + \lambda_{\text{supcon}}(\nu) \mathcal{L}_{\text{supcon}} + \lambda_{\text{ot}}(\nu) \mathcal{L}_{\text{align}} \quad (3-20)$$

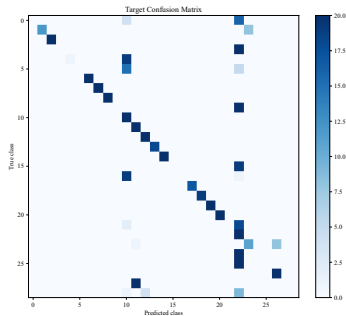
方法一：TPU-DeepJDOT 训练流程

P21

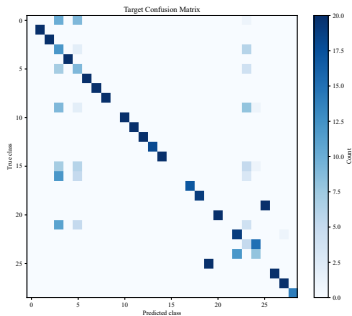


方法一验证: $m_5 \rightarrow m_2$ 对齐递进

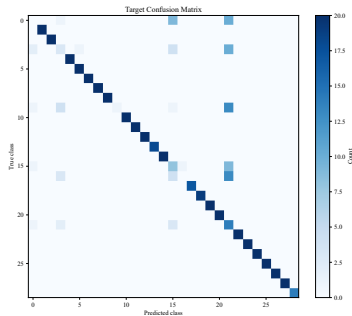
P44



SourceOnly
56.08%



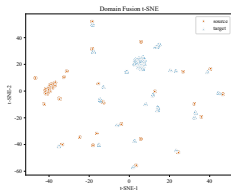
DeepJDOT
67.20%



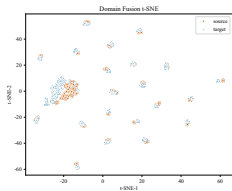
TPU-DeepJDOT
83.60%

方法一验证：域分布与类别结构

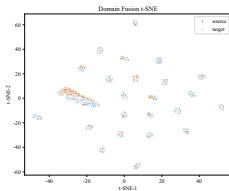
P45



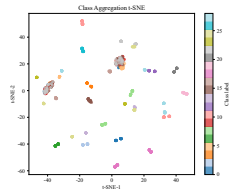
SourceOnly: 域 t-SNE



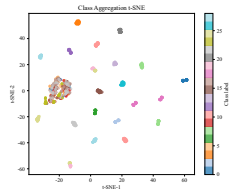
DeepJDOT: 域 t-SNE



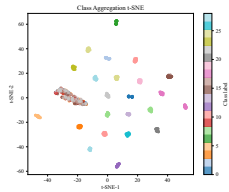
TPU-DeepJDOT: 域 t-SNE



SourceOnly: 类 t-SNE



DeepJDOT: 类 t-SNE

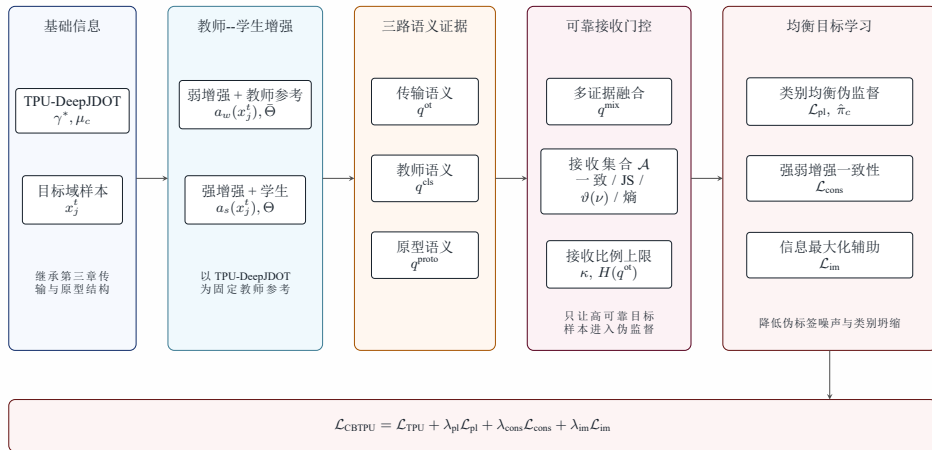


TPU-DeepJDOT: 类 t-SNE

方法二：CBTPU-DeepJDOT 架构

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CBTPU-DeepJDOT 多证据置信均衡伪标签控制机制



证据融合

$$q_{jc}^{\text{mix}} = \frac{(q_{jc}^{\text{ot}})^{\alpha_{\text{ot}}} (q_{jc}^{\text{cls}})^{\alpha_{\text{cls}}} (q_{jc}^{\text{proto}})^{\alpha_{\text{proto}}}}{\sum_{c'=1}^{N_c} (q_{jc'}^{\text{ot}})^{\alpha_{\text{ot}}} (q_{jc'}^{\text{cls}})^{\alpha_{\text{cls}}} (q_{jc'}^{\text{proto}})^{\alpha_{\text{proto}}}} \quad (4-12)$$

接收条件

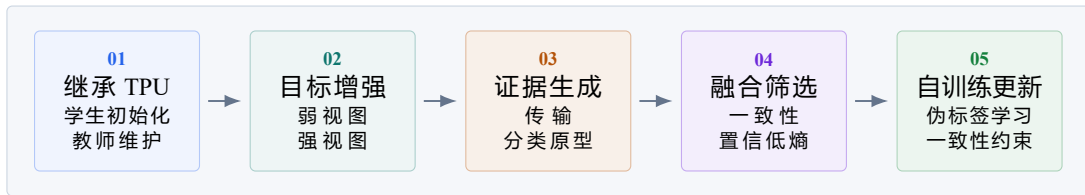
$$\mathcal{A}_j = [\hat{y}_j^{\text{ot}} = \hat{y}_j^{\text{cls}} = \hat{y}_j^{\text{proto}} \text{ or } D_{\text{JS}} \leq \delta_{\text{JS}}] \wedge [\max_c q_{jc}^{\text{mix}} \geq \vartheta(\nu)] \wedge [H(q_j^{\text{ot}}) \leq h_{\text{ot}}] \quad (4-15)$$

总目标

$$\mathcal{L}_{\text{CBTPU}} = \mathcal{L}_{\text{src}} + \lambda_{\text{supcon}}(\nu) \mathcal{L}_{\text{supcon}} + \lambda_{\text{ot}}(\nu) \mathcal{L}_{\text{align}} + \lambda_{\text{pl}}(\nu) \mathcal{L}_{\text{pl}} + \lambda_{\text{cons}}(\nu) \mathcal{L}_{\text{cons}} + \lambda_{\text{im}}(\nu) \mathcal{L}_{\text{im}} \quad (4-22)$$

方法二：CBTPU-DeepJDOT 训练流程

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$$q_{jc}^{\text{mix}}$$

融合证据

$$\mathcal{A}_j$$

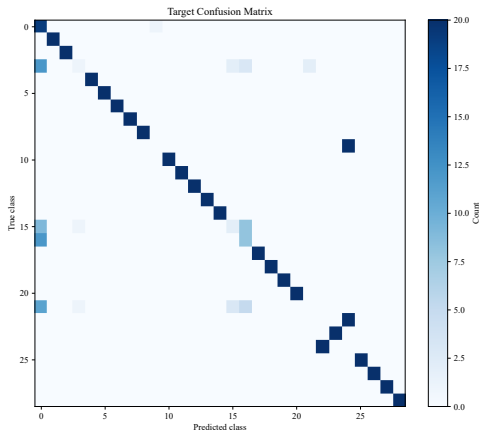
伪标签接收集合

$$\mathcal{L}_{\text{CBTPU}}$$

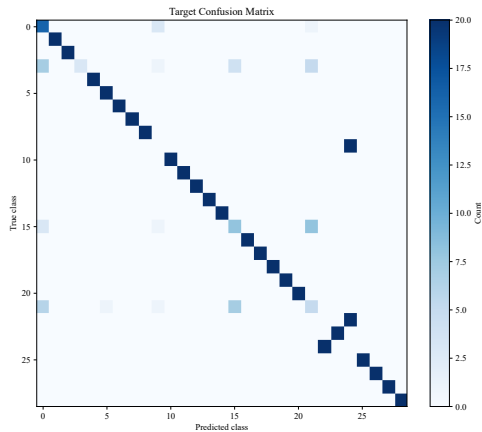
训练目标

方法二验证: $m_5 \rightarrow m_1$ 伪标签细化

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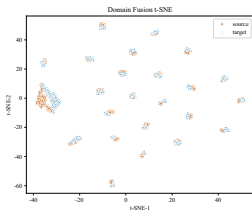
TPU-DeepJDOT
77.59%



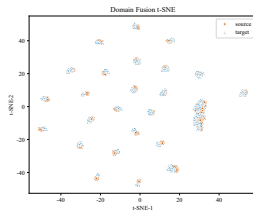
CBTPU-DeepJDOT
81.38%

方法二验证：局部边界继续收紧

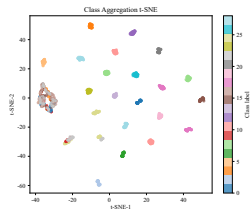
P47



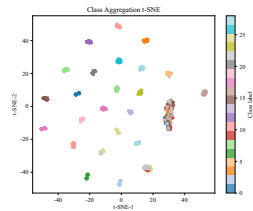
TPU-DeepJDOT：域 t-SNE



CBTPU-DeepJDOT：域 t-SNE

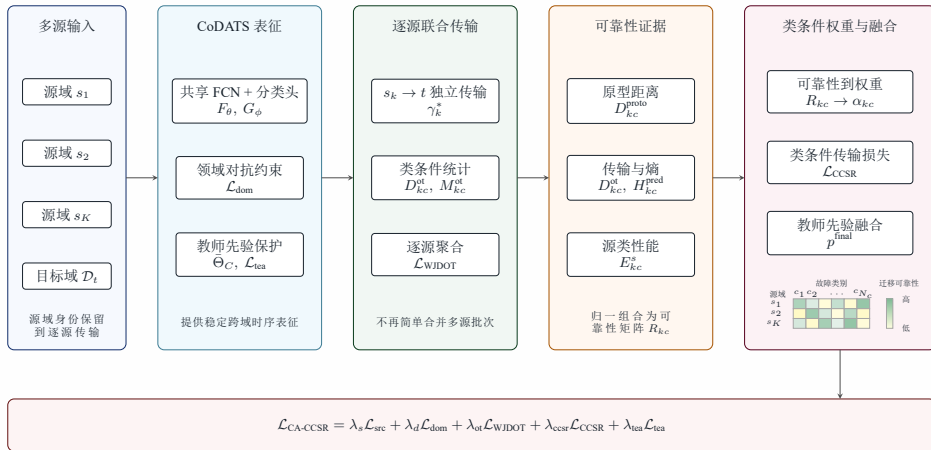


TPU-DeepJDOT：类 t-SNE



CBTPU-DeepJDOT：类 t-SNE

CA-CCSR-WJDOT 类条件源域可靠性加权机制



可靠性评分

$$R_{kc} = w_p \tilde{D}_{kc}^{\text{proto}} + w_o \tilde{D}_{kc}^{\text{ot}} + w_h \tilde{H}_{kc}^{\text{pred}} + w_e \tilde{E}_{kc}^s \quad (5-14)$$

源域权重

$$\alpha_{kc} = \frac{\exp(-R_{kc}/T_{\text{rel}})}{\sum_{k'=1}^K \exp(-R_{k'c}/T_{\text{rel}})} \quad (5-15)$$

CCSR 约束

$$\mathcal{L}_{\text{CCSR}} = \frac{1}{|\mathcal{C}_{\text{act}}|} \sum_{c \in \mathcal{C}_{\text{act}}} \sum_{k=1}^K \mathbb{I}_{kc} \alpha_{kc} D_{kc}^{\text{ot}} \quad (5-17)$$

总目标

$$\mathcal{L}_{\text{CA-CCSR}} = \lambda_s \mathcal{L}_{\text{src}} + \lambda_d(\nu) \mathcal{L}_{\text{dom}} + \lambda_{\text{ot}} r_{\text{ot}}(\nu) \mathcal{L}_{\text{WJDOT}} + \lambda_{\text{ccsr}} r_{\text{ccsr}}(\nu) \mathcal{L}_{\text{CCSR}} + \lambda_{\text{tea}}(\nu) \mathcal{L}_{\text{tea}} \quad (5-22)$$

方法三：CA-CCSR-WJDOT 训练流程

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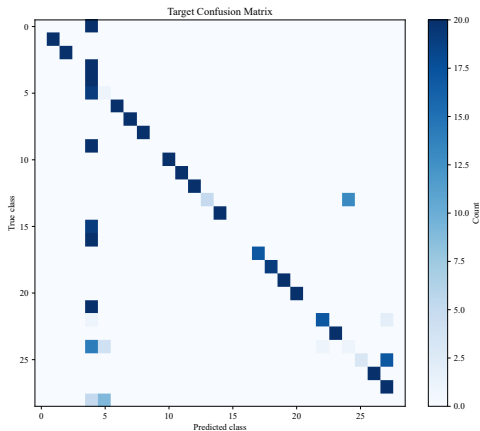
R_{kc}
类条件可靠性

α_{kc}
源域类别权重

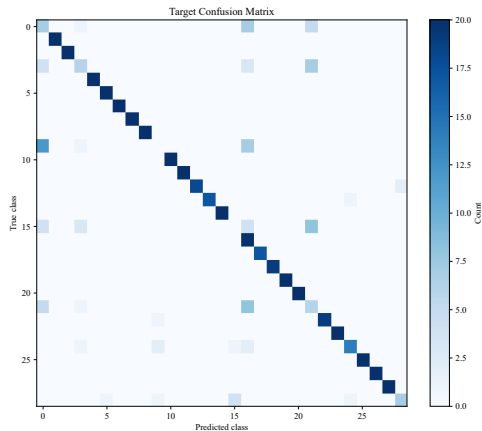
$\mathcal{L}_{\text{CA-CCSR}}$
训练目标

方法三验证：五源代表任务减少误判

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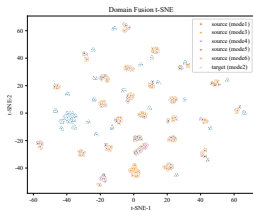
SourceOnly
64.02%



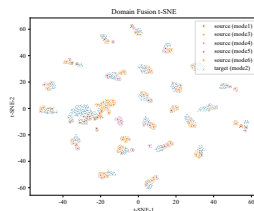
CA-CCSR-WJDOT
82.89%

方法三验证：域间融合与类别聚集

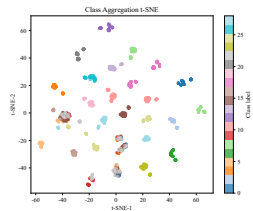
P49



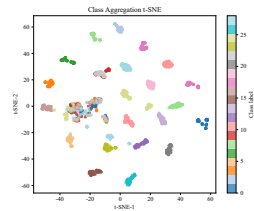
SourceOnly: 域 t-SNE



CA-CCSR-WJDOT: 域 t-SNE



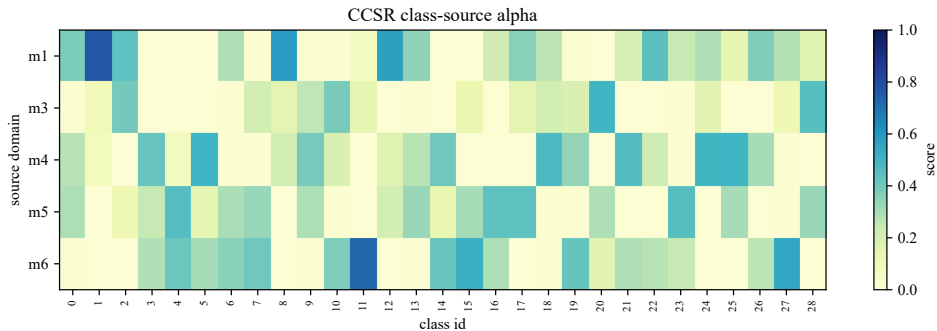
SourceOnly: 类 t-SNE



CA-CCSR-WJDOT: 类 t-SNE

方法三解释：源域-类别权重矩阵

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五源任务 $\{m_1, m_3, m_4, m_5, m_6\} \rightarrow m_2$

源域数	目标域数	场景构造	汇总规模
1	1	m_1, m_2, m_5 全量握手原则	6 单到单场景 × 8 方法
2	1	m_1, m_2, m_5 全量握手原则	3 两源一目标场景 × 5 方法
5	1	6 个工况轮流作为目标	6 五源一目标场景 × 5 方法

 m_1, m_2, m_5 全量握手原则

三个代表工况两两作为源域与目标域；
形成 6 个有向单到单场景；
两源实验对应 3 个留一目标场景。

ACC

整体诊断准确率

主汇总指标

Macro-F1

类别等权 F1

关注少数类

混淆矩阵

类别误判结构

逐任务展示

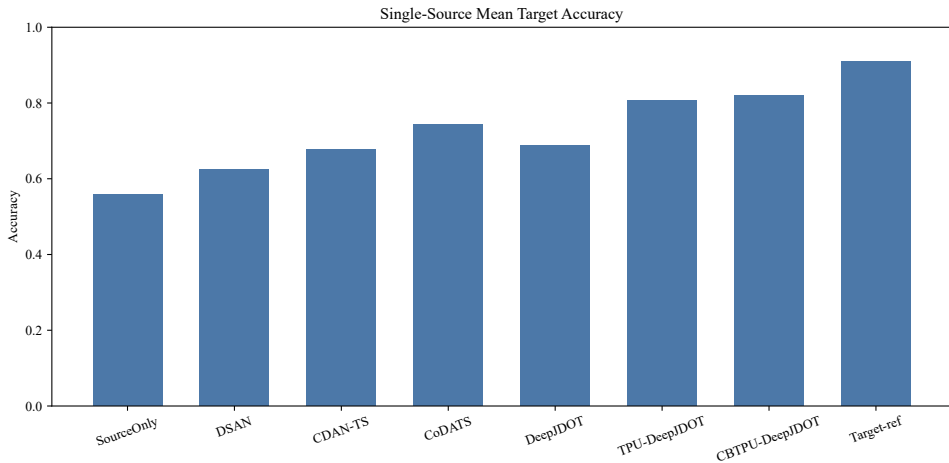
t-SNE

结构证据

看域与类

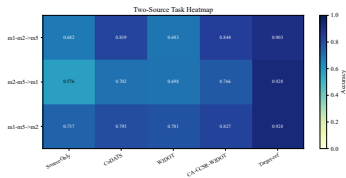
单源结果：主提升来自对齐，细化来自伪标签

P50



多源结果：类别级可靠性压住负迁移

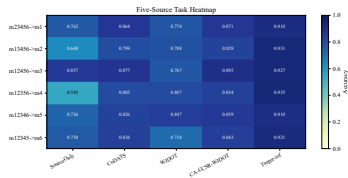
P51



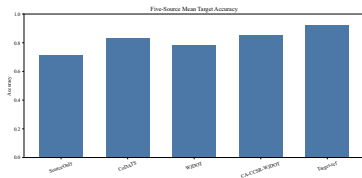
两源任务准确率热力图



两源 CA-CCSR-WJDOT
81.21%



五源任务准确率热力图



五源 CA-CCSR-WJDOT
85.50%

+11.66%

TPU-DeepJDOT $\uparrow$ > DeepJDOT

$p = 0.015625$

+1.48%

CBTPU-DeepJDOT $\uparrow$ > TPU

$p = 0.015625$

+7.69%

CA-CCSR-WJDOT $\uparrow$ > WJDOT

$p = 0.001953$

单源场景中，TPU-DeepJDOT 与 CBTPU-DeepJDOT 相对基线依次提升，体现了对齐约束与伪标签细化的递进作用。

多源场景中，CA-CCSR-WJDOT 相对 WJDOT 基线胜出，说明类别级源域可靠性能够缓解负迁移。

整体上，三套方法在对应实验场景中均取得有效提升，完成了跨生产模式故障诊断课题任务。

谢谢